SMAI Assignment 4 Q1

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Hyperparameter Tuning Approach & Wandb Results :

To systematically search for optimal multi-task model, I have divided experiments into four hyperparameter groups:

Group	Parameters Varied
G1 - Lambda Weights	(λ_1, λ_2) values → [1:1, 2:1, 1:2, 0.5:1]
G2 - Dropout & Learning Rate	dropout → [0.2, 0.3, 0.4], lr → [1e-3, 5e-4, 1e-4]
G3 - Optimizers	Optimizer → [Adam, AdamW, SGD]
G4 - Architecture & Batch size	NF = [16, 64], batch sizes = [32, 64, 128]

Each run was logged using wandb with tracking of :

- Training & Validation loss curves (Total Loss, Cross entropy Loss, MSE Loss)
- Accuracy , MAE , MSE metrics
- Final results dashboard grouped by Experiment category

These visualizations helped quickly to observe

- Which hyperparameters improve classification vs regression
- Trends in task trade-offs

Final Best Models Selection :

After completion of grouped experiments, I selected the two best-performing models by choosing top hyperparameters from each group and rerunning them with best combination:

Model Name	Goal	Final Best Configuration
best_accuracy_model	Maximize class prediction(Accuracy)	λ_1=2.0, λ_2=1.0, NF=64, dropout=0.2, batch=64, lr=1e-3, AdamW
less_rmse_model	Optimize ink usage estimation	λ_1=1.0, λ_2=2.0, NF=64, dropout=0.3, batch=128, lr=1e-3, AdamW



These settings reflect the best hyperparameters learned from each experiment group rather than individual results.

Feature Map Visualization :

I have generated detailed feature map visualization for 5 trained models,
each with :

- Three different test images
- Eight visualized features per encode block
- Total : 120 feature map plots

The model uses three convolution blocks (Enc1 , Enc2, Enc3).

Layer Wise Interpretation

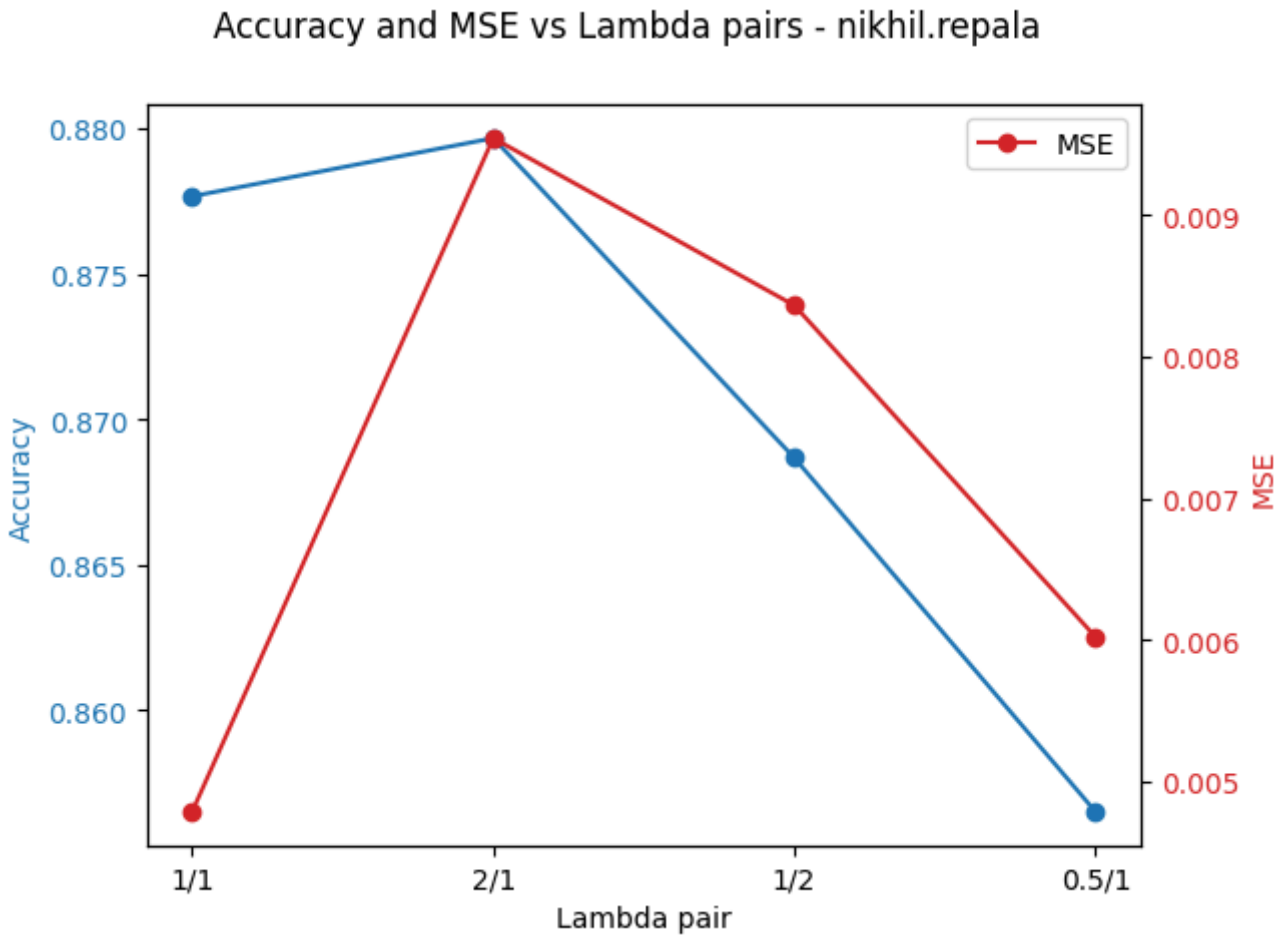
- Encoder Block 1 :
  - Reacts strongly to simple edges and intensity transition boundaries
  - It highlights the outer structure of object
  - Still looks close to the original image (spatial detail is preserved)
  - Helps classification by distinguishing basic silhouettes
  - Helps ink regression by identifying where ink is concentrated at edges
- Encoder Block 2 :
  - Noise and Background starts to disappear
  - The model begins to detect texture patterns and shapes inside object
  - Individual regions (like thin or thick parts) become more prominent
  - Improves category separation among similar items (eg. trousers vs jeans)
  - Provides better signal about region-wise ink density and fabric texture
- Encoder Block 3 :
  - The feature maps now look blocky and very abstract
  - Spatial details are gone
  - The model focuses on the parts that are most useful for decision making
  - Critical for final classification
  - Helps ink regression by estimating total area / silhouette ink coverage

Block	Feature Type Learned	Spatial Detail	Semantic Meaning	Primarily Helps With
enc1	Edges, simple textures, contrast patterns	High	Low	Initial feature extraction for both tasks
enc2	Local clothing textures, folds, regional shapes	Medium	Medium	Ink estimation (pixel intensity regions) & basic class clues



enc3	Overall garment structure, semantic shapes	Low	High	Category prediction & contextual ink reasoning

Task Trade-Off Analysis with $\lambda_1 : \lambda_2$



The experiment clearly shows :

High λ_1 (CE priority)	**High λ_2 (MSE priority)**
Better classification	Better regression
RMSE increases	Accuracy drops
Focus on semantic identity	Focus on pixel intensity magnitude

There is a trade-off between the two objectives — higher λ_1 favors classification accuracy, while higher λ_2 improves regression MSE. The **1:1 ratio** provides a good overall balance for this multi-task CNN.